

Talk Abstract

Cosmic rays are a major environmental hazards to space craft in interplanetary and near-Earth space. Radiation hardness is an important cost driver for space missions. These particles are energetically coupled to the boundaries of our heliosphere, filtered or accelerated by the shock structures. The shape and structure of the heliosphere and all astrospheres are poorly constrained. There are few in-situ measurements of heliosphere boundaries like the termination shock, heliopause, and bow shock. This program will review a multi-fluid MHD model for the heliosphere informed by interstellar probe measurements as well as optical observations from space telescopes, a new inclusion. Our ability to predict the behavior of energetic cosmic rays depends on our knowledge of the heliosphere boundary. Understanding and constraining the behavior of energetic particle interactions in the heliosphere is critical for the future of space exploration.

A diagram of the different species of energetic particles in the heliosphere. **Heliospheric Energetic Particles represent the main radiation hazard for spacecraft.** Cosmic rays originate both inside and outside of the heliosphere. The shock structure on the edges of the stellar wind bubble accelerates and filters cosmic rays. Understanding and constraining the behavior of energetic particle interactions in the heliosphere is critical for the future of space exploration.

